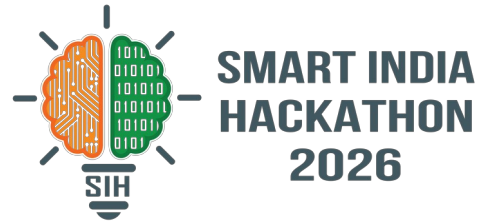


SMART INDIA HACKATHON 2026



NISHAN-PQ

Accountable decryption.
Verifiable source-copy evidence.

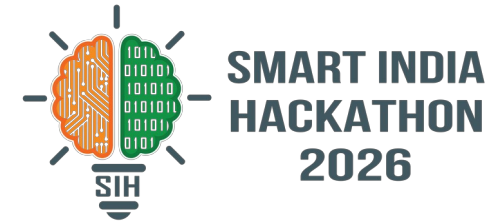
When a protected copy leaks, connect it to a signed release session
—or return inconclusive.

PROBLEM STATEMENT ID	SIH26237
PROBLEM STATEMENT TITLE	Cryptographic Attribution and Immutable Decryption Provenance for Multi-Recipient Encrypted Document Distribution
THEME	Blockchain & Cybersecurity
PS CATEGORY	Software
TEAM ID	[ENTER TEAM ID]
TEAM NAME (REGISTERED ON PORTAL)	[ENTER REGISTERED TEAM NAME]



Evidence status · 12 September 2026

IDEA TITLE / PROPOSED SOLUTION



[TEAM NAME]

One encrypted source. Signed, session-specific releases.

01 Session-specific release

Serialized row + session allocation.
Marked output is signed before publication.

02 Commit before release

ML-DSA receipt enters the local 3-of-4 log. Configured mode checks signed witness state under a caller-provisioned key pin.

03 Conservative trace

Clean PDFs need two-channel agreement. Missing evidence produces a lead or inconclusive result.

CONTROLLED RELEASE → EVIDENCE

Encrypt once
AES-256-GCM



Wrap key
ML-KEM-768



Mark
visual + layout



Sign + commit
ML-DSA-65



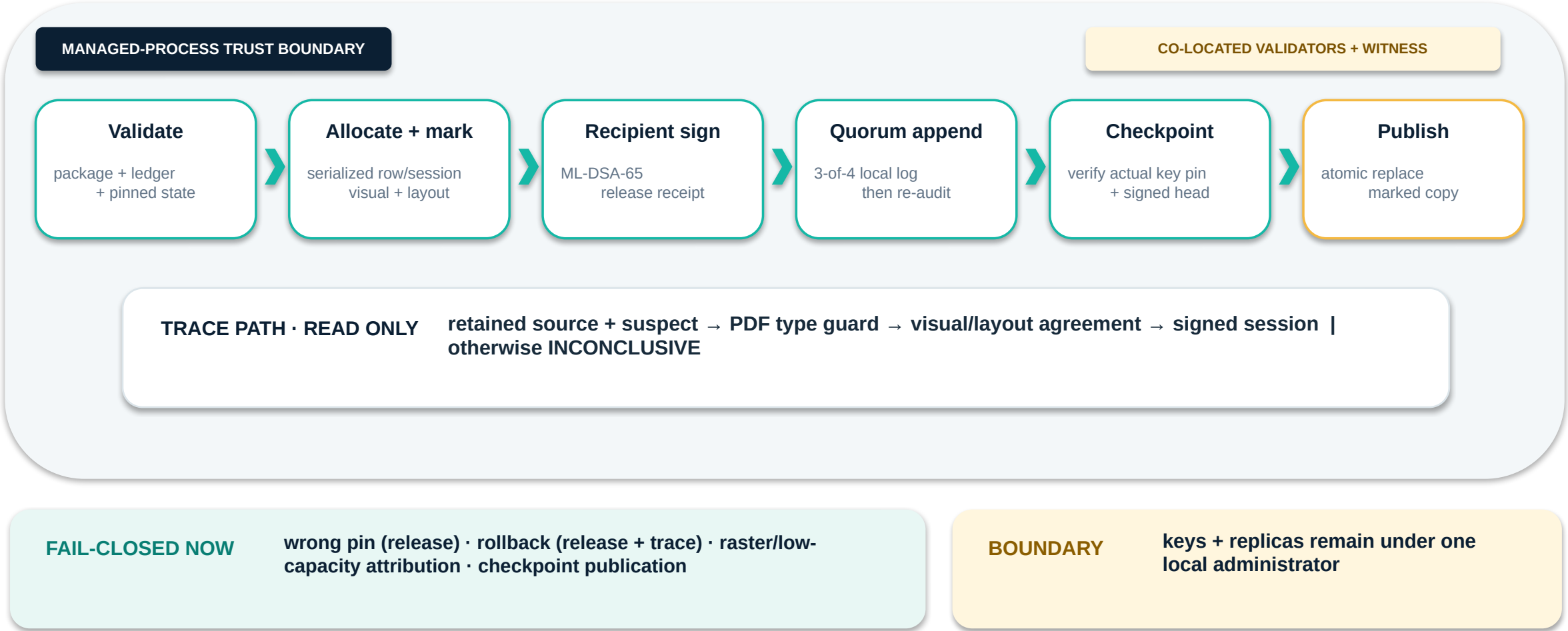
Trace
attribute / abstain

Outcome: verifiable source-copy evidence—not a claim about which human disclosed the file.

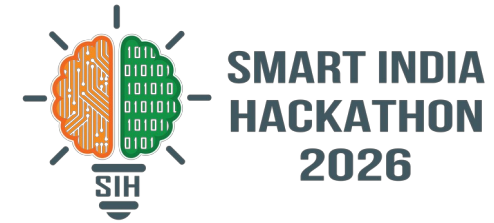
TECHNICAL APPROACH

[TEAM NAME]

Commit and witness before release (configured mode)



FEASIBILITY AND VIABILITY



[TEAM NAME]

Fresh safeguards first; historical visual results and physical evidence stay separate.

FRESH · CURRENT CODE

12/12

safeguard scenarios passed
59/59 final-code tests · real PQC · 19 hashes
verified

- Serialized release: concurrent rows differ; next release succeeds
- Pinned mode: wrong pin blocks release; rollback blocks release + trace
- Checkpoint failure publishes nothing; existing output is preserved
- Clean PDF corroborates; raster and low-capacity PDF abstain

Six-person team · existing offline laptop prototype · repeatable saved evidence

SYNTHETIC OFFLINE SCENARIOS · NOT POPULATION EXPERIMENTS

HISTORICAL VISUAL FIXTURES

30/30 JPEG-Q55 visual-channel trials

One document/codebook; repeated sessions. 5 one-page PDFs retained text; mean PSNR 41.87 dB.

Physical recovery remains experimental: 0/4 at the shipped threshold

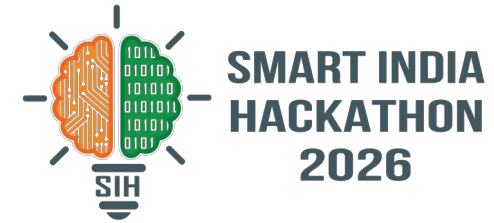
Exploratory profiles recover the same 1/4 capture. No added capture; no robust physical claim.

Digital PDF corroboration; raster-only matches are research leads

IMPACT AND BENEFITS

[TEAM NAME]

A five-minute judge demo shows both the useful answer and the refusal to overclaim.



JUDGE-DRIVEN DEMO · OFFLINE LAPTOP

- | | | |
|----|------------------|--|
| 01 | Issue | Alice + Bob + repeated session → distinct rows and signed copy hashes |
| 02 | Associate | Clean PDF → two channels agree on one signed source copy |
| 03 | Abstain | Rendered transplant / short PDF → inconclusive; research lead retained |
| 04 | Block | Wrong pin blocks release; rollback blocks release + trace; checkpoint fail publishes nothing |

VALUE: A DEFENSIBLE SESSION LEAD—OR A DEFENSIBLE INCONCLUSIVE

OPERATIONAL BENEFIT

- Air-gapped workflow; no cloud KMS or public chain
- Evidence tied to a release session, not a vague access log
- Explicit refusal when high-assurance evidence is missing

NARROW SECOND-ROUND ROADMAP

- Independent administrator + key custody
- Larger multi-recipient, multi-page corpus
- Representative held-out print / camera tests
- Portable independent evidence verifier

No GPU or ESP32 required for the measured demo.

RESEARCH AND REFERENCES

[TEAM NAME]

PRIMARY SOURCES & PRIOR ART

- 1 Official SIH26237 problem statement**
sih.gov.in/sih2026PS
- 2 NIST FIPS 203 / 204 — ML-KEM / ML-DSA**
csrc.nist.gov/pubs/fips/203/final · [/fips/204/final](https://fips/204/final)
- 3 FontCode — established glyph perturbation**
cs.columbia.edu/cg/fontcode/
- 4 PQFabric — prior permissioned PQ ledger work**
[arXiv:2010.06571](https://arxiv.org/abs/2010.06571)
- 5 Tardos fingerprint codes**
renyi.hu/~tardos/fingerprint.pdf
- 6 Saved fresh + historical evidence**
[demo/results.json](#) · [artifacts/nishan/*.json](#) · [scores.json](#)
- 7 Public repository**
[INSERT PUBLIC REPOSITORY LINK BEFORE UPLOAD]

DIFFERENTIATED INTEGRATION

ESTABLISHED COMPONENTS

PQC standards, traitor-tracing codes, document marks and permissioned ledgers all have prior art.



ENGINEERING CONTRIBUTION

Tested integration of PQ-signed marked release, configured pinned-checkpoint gating and conservative two-channel PDF evidence.

INTEGRATION CLAIM · NOT A FIRST-EVER CLAIM